

METAMORPHIC ROCKS

When existing rocks are subjected to heat or pressure or both within the Earth, they are changed into different assemblages of minerals. Contact metamorphism occurs when intense, localized heat radiating from a body of igneous rock causes the surrounding rocks to recrystallize. Regional metamorphism takes place over wide areas, often at considerable depth, as a result of intense heat and pressure. Movements of Earth's crust can result in rocks being pulverized by dynamic metamorphism.



MYLONITE

When rocks are pulverized deep within a thrust fault zone, the rock dust and fragments produced form a fine-grained rock called mylonite.



GARNET SCHIST

The red garnets in this variety of schist indicate that it developed at relatively high temperatures and pressures deep within the continental crust.



BIOTITE SCHIST

Formed at relatively high pressures and temperatures, biotite schist contains feldspars, quartz, and much dark-coloured biotite mica.



SKARN

Formed when rocks rich in carbonate minerals are altered by high-temperature contact metamorphism, skarn contains minerals rich in calcium, magnesium, and iron.

FULGURITE

When lightning strikes in deserts or on beaches, sand can melt, forming small, tube-shaped structures, called sand fulgurites.



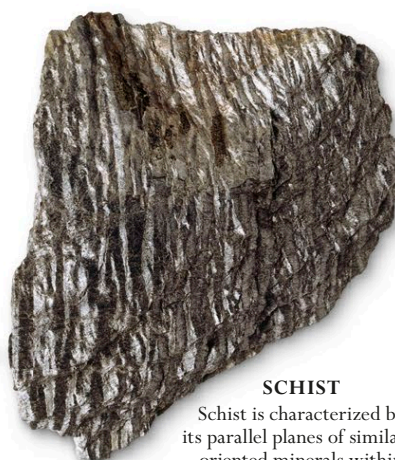
GRANULITE

Formed at very high temperatures and pressures, granulite is dark, coarse-grained, and rich in pyroxenes, garnets, micas, and feldspars.



MUSCOVITE SCHIST

A typical schist with pale, glittery muscovite mica, this rock also contains quartz and feldspars.



SCHIST

Schist is characterized by its parallel planes of similarly oriented minerals within the rock. The corrugated pattern seen on the specimen is called crenulation.



HALLEFLINTA

Originally volcanic tuff, rhyolite, or quartz porphyry, halleflinta is fine-grained, pale-coloured, and rich in quartz. It is a type of hornfels.



SPOTTED SLATE

This is a dark, fine-grained rock characterized by black spots (porphyroblasts) of minerals, such as cordierite and andalusite.



SLATE

Dark and very fine-grained, slate is a compact rock with parallel cleavage planes. It is formed by low-pressure metamorphism.



PHYLLITE

Phyllite is formed at lower temperatures and pressures than schist, but higher than slate. It is fine-grained and splits into slabs with a characteristic sheen.



METAQUARTZITE

With a high percentage of quartz, this rock is harder than most metamorphic rocks. It is formed from sandstone altered at high temperature.

